

MACHINE LEARNING

LINEAR MODEL SELECTION AND REGULARIZATION

Sebastian Engelke

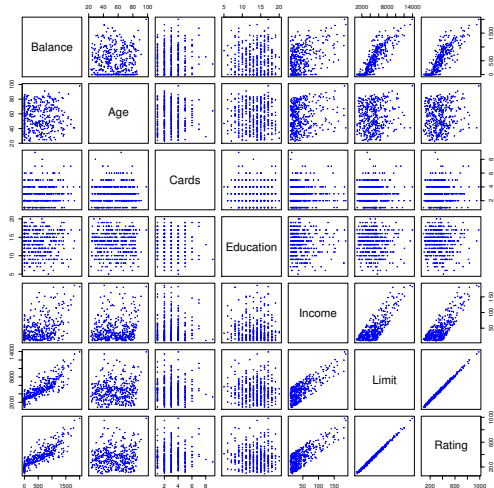
MASTER IN BUSINESS ANALYTICS



**UNIVERSITÉ
DE GENÈVE**

Example: Credit data set

The **Credit data set** from package **ISLR** contains the response **balance** (average credit card debt) and quantitative predictors such as **age**, number of credit **cards**, etc. There are also four qualitative predictors **gender**, **student**, **status** and **ethnicity**; see [ISL], Section 3.3 for treatment of qualitative predictors in linear models.



Data standardization

- ▶ The predictors $X_1, \dots, X_p$ might have very **different scales**.
- ▶ This is not problematic in methods where the parameters can account for these different scales; e.g., linear regression with least squares, LDA, logistic regression.
- ▶ In methods that treat all **predictors equally**, this might cause a problem; e.g., k -Nearest-Neighbor since it uses the distances between predictors x_i and x_j .

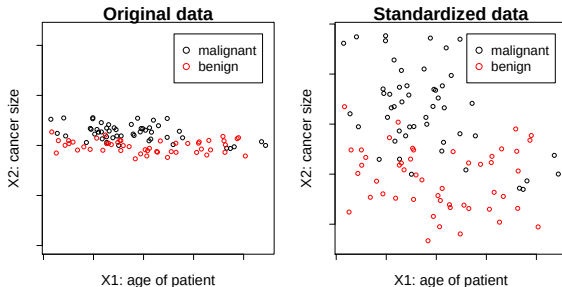
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- ▶ In general, it is good to **standardize the predictors** in the training data x_i , $i = 1, \dots, n$, that is, for each margin $j = 1, \dots, p$:

$$x_{ij}^{\text{scaled}} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}} = \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j},$$

which can be done with the transformer `StandardScaler()` in `python`.

- ▶ For a new input $x_0 \in \mathbb{R}^p$, **standardize margins** first to $x_{0j}^{\text{scaled}} = (x_{0j} - \bar{x}_j) / \hat{\sigma}_j$ and then use the predictive model.



Back to linear models

- ▶ In the **regression setting** a multiple linear model

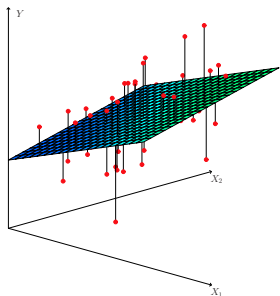
$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p,$$

is fitted to the data (x_i, y_i) , $i = 1, \dots, n$.

- ▶ The estimated vector $\hat{\beta} = (\hat{\beta}_0, \dots, \hat{\beta}_p)$ minimizes the residual sum of squares (RSS)

$$\text{RSS} = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2.$$

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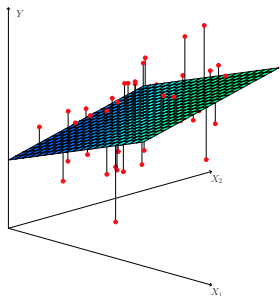
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- (1) **Prediction accuracy**: If n is not much larger than the number of predictors p , or even $p \gg n$ (**high-dimensional setting**), then the least squares estimate $\hat{\beta}$ may have high variance, resulting in overfitting. By **shrinking/regularizing** the estimate we can reduce the variance and improve prediction accuracy on new test data.



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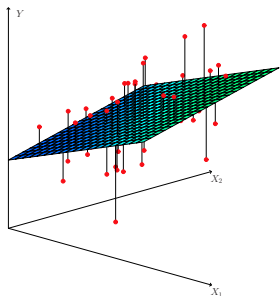
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- (2) **Model interpretability**: Some or many of the predictors might not be associated with the response, but least squares will almost never estimate their coefficients to zero. Excluding these **irrelevant predictors** reduces the complexity of the model and allows interpretation.



Alternatives to least squares

- (1) **Subset selection**: These approaches identify a subset of the p predictors and then fit a linear model by least squares:
- ▶ best subset selection;
 - ▶ stepwise selection (forward/backward)
- ▶ Details in **[ISL], Chapter 6**.

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- (3) **Shrinkage**: A linear regression is fitted to all p predictors, but the estimated coefficients are **shrunk/regularized** towards zero relative to the least squares fit. Depending on the type of regularization (Lasso, Ridge, etc.), some coefficients may be estimated to be exactly zero, and thus shrinkage can perform **automatic variable selection**.

Ridge regression

- ▶ Instead of the RSS the **ridge regression coefficient estimates** $\hat{\beta}_{\lambda}^R$ minimize the penalized version

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2,$$

where $\lambda \geq 0$ is called the **tuning parameter**.

- ▶ The penalty term $\lambda \sum_j \beta_j^2$ is called a **shrinkage penalty** and it is small when $\beta_1, \dots, \beta_p$ are close to 0.

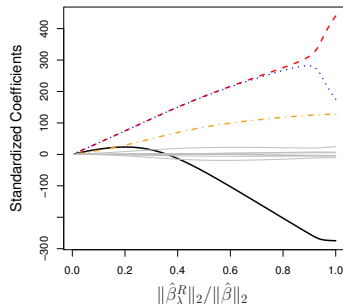
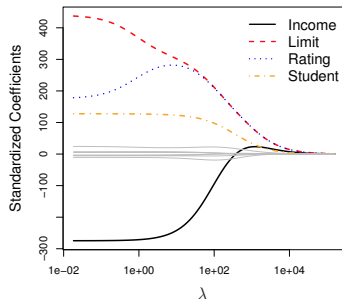
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- ▶ The penalty term $\lambda \sum_j \beta_j^2$ is called a **shrinkage penalty** and it is small when $\beta_1, \dots, \beta_p$ are close to 0.
- ▶ For each $\lambda \geq 0$, ridge regression produces a different set of coefficient estimates $\hat{\beta}_{\lambda}^R$: if $\lambda = 0$ they approach the least squares estimates, and if $\lambda \rightarrow \infty$ they shrink to 0.
- ▶ A good value of λ is critical and is chosen by **cross-validation**.



Ridge regression: closed-form solution

- Ridge regression has a closed-form solution:

$$\hat{\beta}_{\lambda}^R = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T y,$$

where $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the data matrix, $\mathbf{I} \in \mathbb{R}^{p \times p}$ is the identity matrix and $y = (y_1, \dots, y_n)$ is the vector of responses.

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- ▶ We can compare this to the **least squares solution**

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y.$$

- ▶ This shows how ridge regression **shrinks** the least squares estimates to zero.

Lasso: least absolute shrinkage and selection operator

- ▶ Ridge regression has the disadvantage that it includes **all p predictors** in the final model.
- ▶ The **lasso** overcomes this problem. The lasso coefficients $\hat{\beta}_{\lambda}^L$ minimize the penalized RSS

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|,$$

where $\lambda \geq 0$ is again a **tuning parameter**.

- ▶ Only difference to ridge regression: the **lasso penalty** uses the ℓ_1 -norm of β .

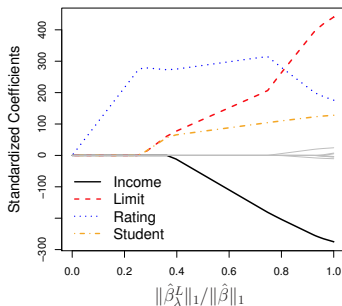
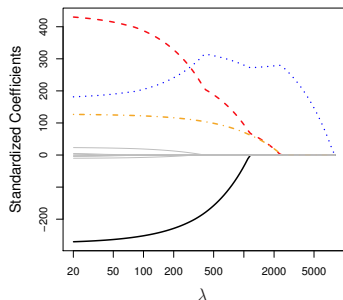
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- ▶ Only difference to ridge regression: the **lasso penalty** uses the ℓ_1 -norm of β .
- ▶ This penalty forces some coefficient estimates to be **exactly 0**, if λ is large enough; lasso thus performs variable selection!
- ▶ Again, a good value of λ is critical and is chosen by **cross-validation**.



Regularization: geometric interpretation

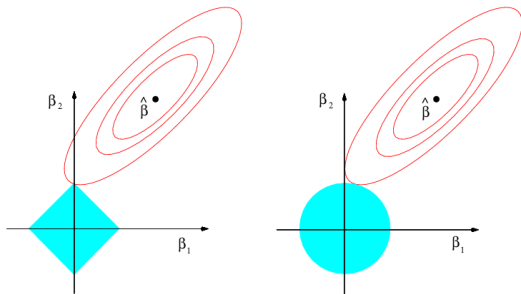
- **Equivalent formulation:** one can show that ridge regression and lasso coefficient estimates $\hat{\beta}_{\lambda}^R$ and $\hat{\beta}_{\lambda}^L$ minimize for some s , respectively,

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s,$$

and

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s.$$

- This allows a geometric interpretation of the effect of **penalization**.



Comments

- ▶ For large p , best subset selection is computationally infeasible. In this case, lasso can be used to perform **variable selection**.
- ▶ While the ridge coefficients $\hat{\beta}_{\lambda}^R$ have a closed-form solution, the lasso coefficients $\hat{\beta}_{\lambda}^L$ do not have a closed-form solution. They are obtained by **numerical optimization**, which is not too hard since the problems are **convex**.
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- ▶ **Elastic-net** (Zou & Hastie, 2005) combines ridge and lasso penalties into

$$\lambda \sum_{j=1}^p \left((1 - \alpha) \beta_j^2 + \alpha |\beta_j| \right),$$

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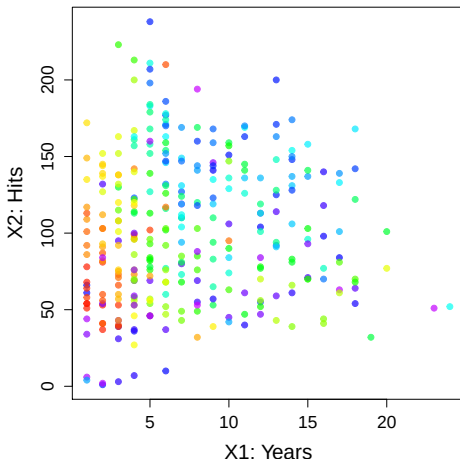
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- ▶ In practice, lasso allows to **pre-select** a reasonable number of variables that are then used in more flexible prediction models.
- ▶ Regularization can be used for other models such as **logistic regression**, etc.

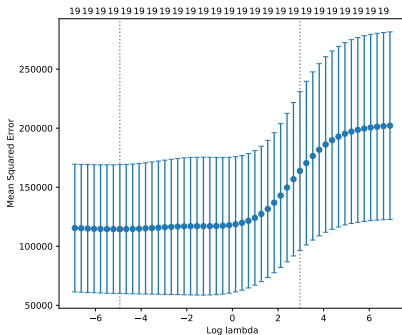
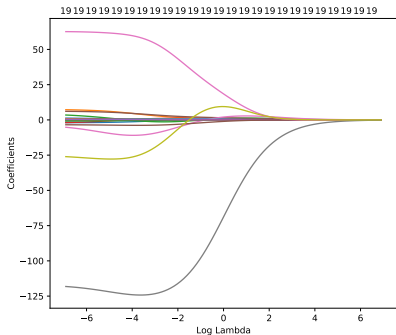
Example: Hitters data set

The `Hitters` data set from the `ISLR` package contains the annual `Salary` of 263 baseball players and $p = 19$ predictors such as numbers `Years` played in major leagues and the number of `Hits`. `Salary` color-code: low (yellow-red) to high (blue, green).



Ridge regression in python

```
from sklearn.linear_model import Ridge
from sklearn.model_selection import GridSearchCV, KFold
X, y = pd.get_dummies(data, drop_first=True).drop("Salary", axis=1), data["Salary"]
grid = 10 ** np.linspace(-3, 3, 100)
coefs = []
for a in grid:
    ridge = Ridge(alpha=a, normalize=True).fit(X, y)
    coefs.append(ridge.coef_)
ridgeCV = GridSearchCV(Ridge(normalize=True), {'alpha':grid}, scoring="neg_mean_squared_error",
                        cv=KFold(n_splits=10, shuffle=True, random_state=1)).fit(X, y)
resCV = ridgeCV.cv_results_
best_ind = np.argmin(-resCV["mean_test_score"])
one_std_best = np.max(grid[-resCV["mean_test_score"] <= \
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```



Lasso in python

```
from sklearn.linear_model import Lasso
from sklearn.model_selection import GridSearchCV, KFold
X, y = pd.get_dummies(data, drop_first=True).drop("Salary", axis=1), data["Salary"]
grid = 10 ** np.linspace(-2.6, 1.7, 100)
coefs = []
for a in grid:
    lasso = Lasso(alpha=a, normalize=True, max_iter=1e5).fit(X, y)
    coefs.append(lasso.coef_)
lassoCV = GridSearchCV(Lasso(normalize=True, max_iter=1e5), {'alpha':grid}, scoring="neg_mean_squared_error",
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